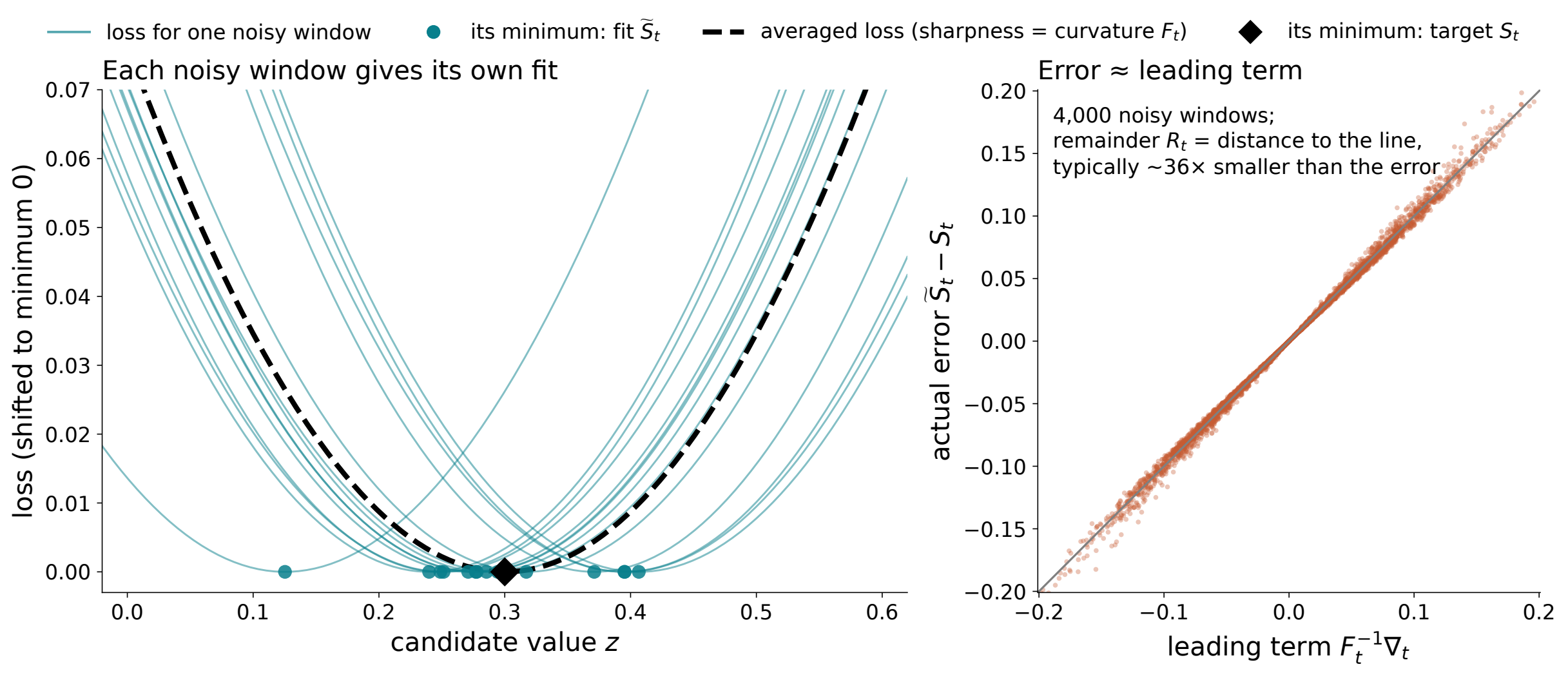

loss for one noisy window
its minimum: fit $\widetilde{S}_t$
averaged loss (sharpness = curvature $F_t$)
its minimum: target $S_t$
Each noisy window gives its own fit
loss (shifted to minimum 0)
candidate value $z$
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Error ≈ leading term
4,000 noisy windows;
remainder $R_t$ = distance to the line,
typically ~36× smaller than the error
actual error $\widetilde{S}_t - S_t$
leading term $F_t^{-1}\nabla_t$
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
−0.2
−0.1
0.0
0.1
0.2